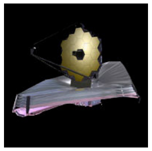

9294 - Accretion-induced X-ray emission from the closest candidate polar

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Tim Cunningham (PI)	**Harvard University**
Dr. Ilaria Caiazzo (CoI) (ESA Member)	Institute of Science and Technology Austria
Prof. Peter J. Wheatley (CoI) (ESA Member)	University of Warwick
Prof. Boris T. Gaensicke (CoI) (ESA Member)	University of Warwick
Dr. Pasi J. Hakala (CoI) (ESA Member)	University of Turku
Joseph Alexander Guidry (CoI)	Boston University
Dr. JJ Hermes (CoI)	Boston University
Dr. Scott J. Kenyon (CoI)	Smithsonian Institution Astrophysical Observatory
Prof. David Charbonneau (CoI)	Harvard University
Dr. Kishalay De (CoI)	Columbia University in the City of New York
Dr. John Charles Raymond (CoI)	Smithsonian Institution Astrophysical Observatory
Mr. Sumit Kumar Maheshwari (CoI) (ESA Member)	Universitat Hamburg

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(1) ATO-J026.6489+49.2452
	2	MIRI imaging	MIRI Imaging	(1) ATO-J026.6489+49.2452

ABSTRACT

A large fraction of the population of Galactic cataclysmic variables (CVs) should be made of "period bouncers", i.e.systems that have evolved past the well-known period minimum and are now evolving toward longer periods. We here propose to observe the most interesting of such systems, ZTF

J0146+4914. This is the closest polar to Earth, and it is the most highly magnetized, longest-period period-bouncer known, making it possibly the most evolved and oldest CV known. At only 56 pc, this is an exceptional target for multi-wavelength X-ray/IR observations. The proposed Chandra and JWST observations will allow us to measure the accretion rate and characterize the properties of the ultracool substellar donor in the system, and therefore provide an important benchmark for the study of the evolution of CVs.

OBSERVING DESCRIPTION

A large fraction of the population of Galactic cataclysmic variables (CVs) should be made of "period bouncers", i.e.systems that have evolved past the well-known period minimum and are now evolving toward longer periods. We here propose to observe the most interesting of such systems, ZTF J0146+4914. This is the closest polar to Earth, and it is the most highly magnetized, longest-period period-bouncer known, making it possibly the most evolved and oldest CV known. At only 56 pc, this is an exceptional target for multi-wavelength X-ray/IR observations. The proposed Chandra and JWST observations will allow us to measure the accretion rate and characterize the properties of the ultracool substellar donor in the system, and therefore provide an important benchmark for the study of the evolution of CVs.

JWST is the only facility that can detect and characterize the donor in this system. We request a 2.42-hr observation with NIRSpec (0.4' x 3.8", G395/F290LP) and 52 minutes of exposures with MIRI imaging in 4 filters (F770W/6min, F1280W/9min, F1500W/9min, and F1800W/29min), totalling 5.5 hr including overheads.

We request 100 groups, 7 integrations and 2 dithers for NIRSpec using the SUBS400A1 subarray and NRS readout pattern. This will yield 14 integrations, with total exposure time 2hr\,25m, and ensures that our target will not saturate. The NIRSpec setup targets two key features of the WD+BD system: 1) the white dwarf

continuum between 2.8--3.5 micron, which will improve the constraints on the properties of the accretor, and 2) the brown dwarf CO molecular band feature centered at 4.5 micron. With our choice of exposure time, we will achieve a SNR>50 in the brown dwarf features (after subtracting the white dwarf continuum) for a 500 K brown dwarf, and a SNR>30 for a 300 K brown dwarf.

Furthermore, we have ensured that our total exposure time covers slightly more than one full orbital period (1.2 periods in total) with 14 integrations. This setup will also allow us to investigate time variability of the brown dwarf donor. Even when dividing our observation in 4 phase bins, we will still achieve a high SNR in the single phase-resolved spectra, enough to characterize the slight change in temperature that we might expect because of irradiation.

For MIRI imaging, we request between 20-50 groups, selected for each filter to ensure that the background will not saturate. The MIRI imaging setup is also designed to target two key features: 1) the white dwarf continuum at 7.7 micron, where the contribution of the brown dwarf is expected to be negligible, and 2) the MIR brown dwarf features which include the NH3 feature at 8-15 micron. With our choice of exposures, we will be able to detect with high significance any excess, even in the case the brown dwarf is much colder than expected.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ATO-J026.6489+49.2452	RA: 01 46 35.7258 (26.6488575d) Dec: +49 14 43.09 (49.24530d) Equinox: J2000	Proper Motion RA: 7.248 mas/yr Proper Motion Dec: -8.9520000756238 mas/yr Parallax: 0.0176436" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Cataclysmic variable stars, Circumstellar dust, White dwarfs]
Extended=NO

Observation

Proposal 9294, Observation 1: NIRSpec

Mon Oct 20 21:00:16 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ATO-J026.6489+49.2452	RA: 01 46 35.7258 (26.6488575d) Dec: +49 14 43.09 (49.24530d) Equinox: J2000	Proper Motion RA: 7.248 mas/yr Proper Motion Dec: -8.9520000756238 mas/yr Parallax: 0.0176436" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Cataclysmic variable stars, Circumstellar dust, White dwarfs]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	224778.12

Template

HFF Readout Mode	Slit	Subarray
false	S400A1	SUBS400A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	5	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	S400A1	NRSRAPID	111	5	1	NONE	10	50	8725.824	224778.7

Proposal 9294 - Observation 2 - Accretion-induced X-ray emission from the closest candidate polar

Observation

Proposal 9294, Observation 2: MIRI imaging

Mon Oct 20 21:00:16 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ATO-J026.6489+49.2452	RA: 01 46 35.7258 (26.6488575d) Dec: +49 14 43.09 (49.24530d) Equinox: J2000	Proper Motion RA: 7.248 mas/yr Proper Motion Dec: -8.9520000756238 mas/yr Parallax: 0.0176436" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Brown dwarfs, Cataclysmic variable stars, Circumstellar dust, White dwarfs]
Extended=NO

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4						DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	244704.15
2	F1280W	FASTR1	50	1	1	Dither 1	4	4	555.008	244704.13
3	F1500W	FASTR1	25	2	1	Dither 1	4	8	566.108	244704.20
4	F1800W	FASTR1	25	6	1	Dither 1	4	24	1720.525	244704.23